

RotorLab — Final Validation Report

Rushton turbine stirred tank: steady MRF and transient sliding mesh validated against the published Power Number, following identification and correction of a blade placement defect. $T = 0.300$ m, $D = 0.100$ m, $N = 5$ rev/s, water.

1. Results

Metric	Steady MRF	Sliding Mesh	Benchmark
Torque	0.192562 N·m	0.193983 N·m (time-averaged)	—
Power Number	4.85	4.89	≈5.0
Deviation from benchmark	−3.0%	−2.3%	—
Agreement between methods	—	0.74%	—

Both results sit inside the experimentally reported band for a standard six-blade Rushton disc turbine (≈4.6–6.0 depending on blade thickness and proportions; Rushton et al. 1950, Bates et al. 1963). Sliding-mesh torque was averaged over two complete periods (0.0905–0.1905 s, 180 timesteps); the two periods agree to **0.003%** (0.193980 vs 0.193986 N·m).

2. The Diagnostic Arc

Discrepancy. The as-built case gave $Po = 10.49$ (MRF) and 10.67 (sliding mesh) — both ≈2.1× the benchmark, yet within 1.7% of each other.

First hypothesis, refuted. Steady MRF cannot resolve blade-passage unsteadiness, a documented weakness for radial impellers. Transient sliding mesh was the direct test and landed within 1.7% of MRF rather than moving toward the benchmark. Rejected by its own experiment.

Second hypothesis, rejected on magnitude. Realizable K-Epsilon is documented to over-predict power draw for Rushton geometries, but by roughly 10–30%, not 110%. The explanation could not carry the error.

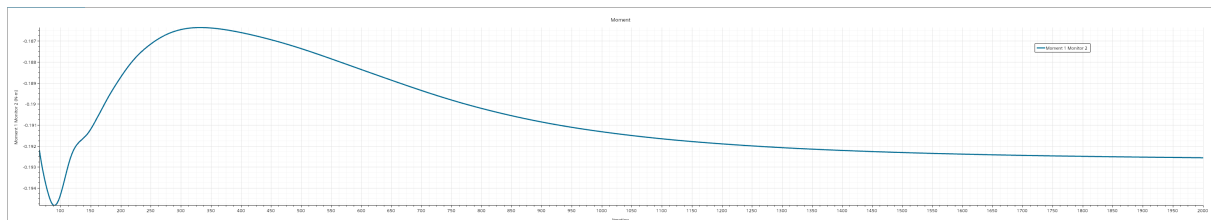
Dimensional audit. Surface area of ImpellerWall (0.0192 m²) matched impeller-only sizing, clearing the torque integration surface. But area is invariant to radial placement. Isolating the blades by Z-threshold and measuring radius directly gave max radius **0.0613 m** against an expected 0.050 m.

Root cause. Blades built as a Block at Base Center [0.0495, 0, 0.100], length 0.026 m — spanning $r = 0.0365 \rightarrow 0.0625$ m instead of $0.025 \rightarrow 0.050$ m, and sitting above the disc plane rather than straddling it. Swept diameter was 0.125 m, not 0.100 m.

Quantity	Value
$(r_2^4 - r_1^4)$ as built (0.0365 → 0.0625)	1.3484×10^{-5}
$(r_2^4 - r_1^4)$ standard (0.0250 → 0.0500)	5.8594×10^{-6}
Predicted torque ratio	2.301
Predicted corrected torque	≈0.199 N·m
Measured corrected torque (MRF)	0.192562 N·m — 3.4% from prediction

Correction applied: Block length 0.026 → 0.025 m, Base Center → [0.0375, 0, 0.090]. Verified post-remesh by Maximum Position[X] on a Z-thresholded slice of ImpellerWall: **0.0500 m exactly**.

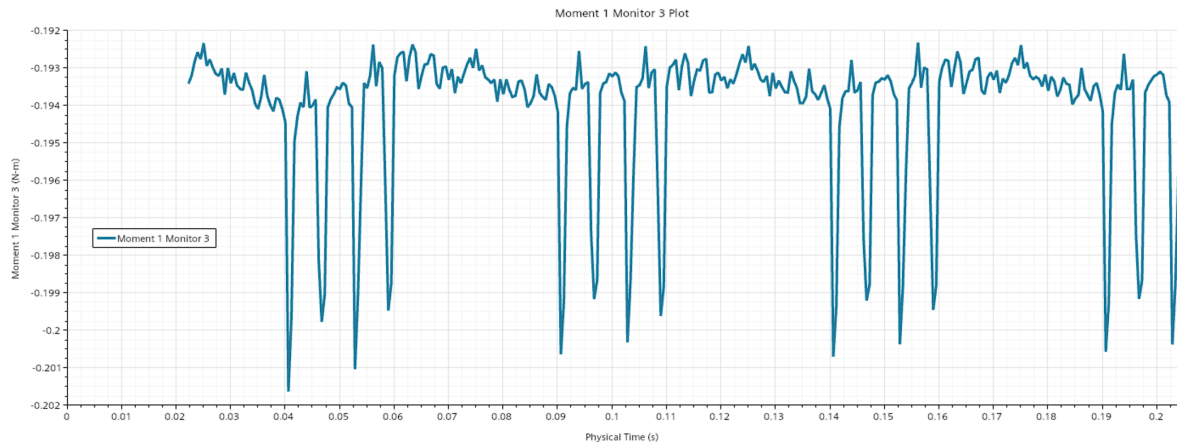
3. Steady MRF Convergence



Torque converged over 2000 iterations to 0.192562 N·m, flat to 0.003% across the final 20 iterations.

Residual at iteration 2000	Value	Residual	Value
Continuity	1.14×10^{-4}	Z-momentum	1.16×10^{-3}
X-momentum	8.08×10^{-4}	Tke	4.01×10^{-5}
Y-momentum	1.20×10^{-3}	Tdr	8.75×10^{-3}

4. Transient Sliding Mesh — Periodic Structure



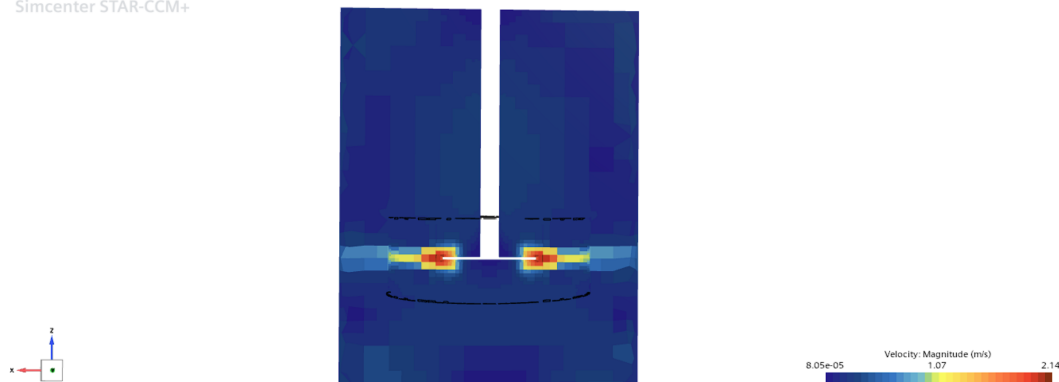
Torque settles into a strictly repeating signal with a 0.0500 s fundamental period — exactly the blade-baffle passing interval for 4 baffles at 0.2 s/revolution. This is the result steady MRF cannot produce.

Cluster	Onset	Interval
1	0.0405 s	—
2	0.0905 s	0.0500 s
3	0.1405 s	0.0500 s
4	0.1905 s	0.0500 s

Each cluster contains the same four-excursion deep-shallow-deep-shallow substructure with matching amplitudes between periods. Baseline $-0.1935 \text{ N}\cdot\text{m}$, excursions to $-0.2007 \text{ N}\cdot\text{m}$ (3.7%).

5. Flow Field

Simcenter STAR-CCM+



Velocity magnitude (Lab Reference Frame), plane section. Radial discharge jet from the blade tips at up to 2.14 m/s splitting into upper and lower circulation loops; both fluid regions fully coupled across the interface.

6. Engineering Findings

A steady-adequate mesh can fail under transient. The corrected mesh converged cleanly under MRF (Continuity 1.14×10^{-4}) but stalled per-timestep under sliding mesh at 10 inner iterations, with Continuity stuck at 4.75×10^{-2} . Diagnostics showed localised degeneracies from the blade-disc Boolean intersection: 93.19° maximum skewness, 15 μm minimum centroid spacing, $3.65 \times 10^{-13} \text{ m}^3$ minimum cell volume. Typical CFL was ≈ 0.7 but ≈ 55 in those cells. A steady solver iterates local stiffness away; a transient solver must resolve it within each timestep. Raising inner iterations to 25 improved end-of-timestep Continuity by **330×**.

3D-CAD primitive conventions. A Cylinder's "Center" is the base profile centre, not the volumetric centre; a Block's "Base Center" is centred in X/Y but at the base in Z. Both produced geometry offset by half a dimension, and both passed 100% mesh face-validity checks.

Diagnostic reach of standard checks. Mesh diagnostics, surface-area checks, and residual behaviour were all clean while the geometry was wrong by 25% in diameter. Only direct dimensional measurement on a thresholded boundary slice exposed it.

7. Limitations

- Sliding mesh run for ≈ 1 revolution (0.2033 s). Justified by periodicity established across four consecutive clusters with 0.003% period-to-period agreement, but a longer run would tighten the average.
- The intra-cluster substructure (≈ 4 excursions at ≈ 0.0062 s spacing) does not map cleanly onto 6-blade, 4-baffle, or 24-encounter frequencies. Reported as observed; no mechanism claimed.
- Mesh quality at the blade-disc intersection is marginal (93.19° max skewness). Adequate for both solutions obtained; repairing it would be the first step in any extension.
- No turbulence-model sensitivity study. With both methods inside the experimental band this is no longer needed to explain a discrepancy, but SST k- ω would be a reasonable robustness check.